

Section E — Threshold Choice and the No-Drift Control

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This section collects results that belong to no single question of the subject and that would distort the one they were attached to. Two are established here. First, detecting a drift and winning the race against the forest are different things, and they separate as soon as the threshold rises. Second, the forest replaces trees at a steady rate when nothing changes at all, which disqualifies the date of the first replacement as a signal on its own. Everything below is recomputed offline from the recorded error trajectories by `resultats_R2/scripts/chiffres_QE.py`; no campaign is rerun.

Control. S_t does not depend on λ , so a single offline pass over each trajectory yields the first crossing for every threshold. The sweep must reproduce the campaign exactly at the three thresholds of the subject, and it does: 1906, 778 and 1 alarms at $\lambda = 8, 25$ and 50. Nothing below is readable without this check.

1 Detecting Is Not Winning

The campaign of 2000 runs is read at six thresholds. The drift-free arm of A1 (2000 runs, no change point at all, same horizon) gives what the same threshold costs when nothing happens.

λ	alarm fires	wins the race	of uncensored	false alarms, no drift
8	95.3%	91.0%	95.4%	83/2 000 = 4.15% [3.36; 5.12]
12	94.3%	78.8%	83.5%	11/2 000 = 0.55% [0.31; 0.98]
15	92.9%	36.3%	39.1%	3/2 000 = 0.15% [0.05; 0.44]
20	66.2%	8.1%	12.2%	0/2 000 = 0.00% [0.00; 0.19]
25	38.9%	2.5%	6.6%	0/2 000 = 0.00% [0.00; 0.19]
50	0.1%	0.0%	0.0%	0/2 000 = 0.00% [0.00; 0.19]

“Wins the race” is $\tau_{\text{det}} < \tau_{\text{ARF}}$ over all 2000 runs, censored runs counted as losses; the next column restricts to the runs that alarm. Brackets: Wilson 95% interval, $n = 2000$ drift-free runs.

The two columns come apart. At $\lambda = 15$ the alarm still fires in 92.9% of runs, barely below the 95.3% of $\lambda = 8$, but it arrives first in only 36.3% of them. Almost two thirds of its alarms are raised after the forest has already replaced a tree — the *Zombie alarms* of the manuscript. Reading the detection column alone would suggest that a moderate threshold escapes the blind spot. It does not.

The paradoxical signature survives at $\lambda = 15$. Amplitude by amplitude, the share of runs won by the detector falls as the drift grows, which is the signature the manuscript reports at $\lambda = 25$.

Δe	$\lambda = 8$	$\lambda = 15$	$\lambda = 20$
0.028	0%	0%	0%
0.085	62%	38%	20%
0.141	88%	81%	49%
0.243	97%	71%	21%
0.361	98%	44%	3%
0.498	99%	18%	1%

At $\lambda = 8$ the rate climbs with the amplitude, from 62% to 99%. At $\lambda = 15$ it peaks at 81% then falls to 18%, and at $\lambda = 20$ it collapses to 1%. The transition $\lambda^* \approx 15$ to 25 announced in the manuscript is therefore confirmed on the race, not contradicted. Q D.1, Table 2 gives the same column at $\lambda = 8$ over all twenty amplitudes.

What the false-alarm column adds. The manuscript grants that a hyper-reactive threshold reaches a safe zone but states that it “sacrifices false-alarm robustness”. On this bench that cost is measured rather than assumed: $\lambda = 8$ raises an alarm in 83 of 2000 runs where nothing changes, 4.15%, just under the 5% calibration level usually adopted for drift detectors. So on the Bernoulli bench the only threshold that wins the race does so at an ordinary false-alarm cost, and the admissible set is not empty here.

Three reservations. The false-alarm rate is measured over 2000 steps on 2000 runs; over a longer deployment such alarms accumulate, and the upper end of the Wilson interval at $\lambda = 8$ is 5.12%. The bench is homoscedastic, so this says nothing about the *Fundamental Tension* the manuscript establishes on ARMA–GARCH streams, where bounding false alarms under clustered volatility is what empties the admissible set. And at the weakest amplitude no threshold wins a single run, which Q A establishes without any dependence assumption.

2 The No-Drift Control

The same drift-free arm answers a question that needs no threshold at all: what does τ_{ARF} read when there is nothing to repair?

On 2000 runs with no change point	
runs where at least one tree is replaced	2000/2000
first replacement, median	95 steps (quartiles 39 and 192)
distinct trees renewed over the horizon, median	9/10
runs whose forest is entirely renewed	575/2000 = 28.7%
median τ_{ARF} at $\Delta e = 0.028$ (real drift)	118 steps
median τ_{ARF} over the whole grid	38 steps

Reading. Replacement churn under pure noise is the normal regime of the ARF, not an anomaly: nine trees out of ten are renewed over two thousand steps with no drift whatsoever. At the weakest amplitude of the grid the first replacement comes after 118 steps against 95 with no drift at all, so at that amplitude it does not distinguish a real change from its absence. This is the control any candidate replacement for τ_{ARF} must pass before its correlations are worth computing, and it is test A1 of the benchmark.

3 What This Section Establishes

Detection and victory separate: a threshold can alarm on almost every drift and still arrive last on two thirds of them, so a detection rate quoted alone does not describe the blind spot. On this bench one threshold does win the race, $\lambda = 8$, at a false-alarm cost of 4.15% per two thousand drift-free steps, which restricts the manuscript’s “no CUSUM threshold escapes it” to the settings it actually tested here. And the date of the first replacement fires as readily without drift as with the weakest one, which is a property of the indicator rather than of the detector.